

John Vandermeulen

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Education

Clemson University

B.S. Mechanical Engineering, Minor in Mathematical Sciences

M.S. Mechanical Engineering (Thermal & Fluid Sciences)

Clemson, SC

Dec 2025 | GPA: 3.48

Dec 2026 | GPA: 3.80

Work Experience

SpaceX

Bastrop, TX

Build Reliability Graduate Engineer

Aug 2026-Dec 2026

- Own qualification testing for V5 automotive Starlink user terminals, identifying production and design failure modes to validate hardware reliability.
- Analyze production processes and hardware to identify potential failure modes and develop qualification test plans.

Gulfstream Aerospace Corporation

Savannah, GA

Cabin Systems R&D Co-op

May 2026-Aug 2026

- Designed PCB schematics and layouts in Altium Designer to support fabrication of aircraft electronic hardware.
- Performed PCB assembly, soldering, electrical troubleshooting, and hardware validation.

Vendor Project Engineering Co-op

May 2025-Aug 2025

- Authored 28 engineering reports supporting vendor airworthiness certification.
- Developed DO-160 qualification test plans and CATIA test fixtures for aircraft components.

Stress Engineer Co-op

Aug 2024-Dec 2024

- Verified FAA Part 25 compliance of 14 G700 interior components through FEMAP analysis and hand calculations.
- Generated stress analyses and certification reports supported by Excel VBA automation.

Baseline Design Engineer Co-op

Jan 2024-Apr 2024

- Designed 37 aircraft interior modifications using CATIA and ENOVIA SmarTeam.
- Standardized G700 galley tower designs using CATIA, incorporating manufacturing feedback from Kaizen walks and floor technicians to improve design reusability.

Skills

- CAD & Design: CATIA, SolidWorks (CSWA), Onshape, AutoCAD, Altium
- Analysis: FEMAP, ANSYS Fluent, MATLAB, XFOIL, Computational Fluid Dynamics, VBA
- Software and PLM: Excel, Word, PowerPoint, SharePoint, ENOVIA SmarTeam
- Training: GD&T, Lean Six Sigma Yellow Belt, Advanced CATIA Training (40 hrs)

Specialized Projects

- 2D Viscous CFD Nozzle Solver: Compressible Flow, MacCormack Scheme, Body-Fitted Grid
- Composite Layup Optimization: Laminate Theory, Tsai-Wu Failure, MATLAB Automation
- Turbojet Engine Optimization: Cycle Analysis, Compressor/Turbine Matching
- Aircraft Design Project: Aerodynamics, Performance, and Stability Sizing
- Clemson Rocketry - Propulsion Sub Team: Hybrid Rocket Engine Design and Simulation